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The Child Penalty Atlas

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1 Big picture

- **Question.** How much of global gender inequality in employment is explained by *child penalties* — the effect of parenthood on women relative to men — and how do these penalties vary across countries, cities, and stages of development?
- **Core contribution.** The paper builds a *world atlas of child penalties* using harmonized microdata from **134 countries**, covering about **7.3 billion people** or **95.5% of the global population**. This is a major expansion relative to the earlier literature, which focused mostly on a small set of high-income countries.
- **Main methodological innovation.** Because panel data are unavailable in most countries, the paper uses a *pseudo-event study* approach that reconstructs pre-child observations from repeated cross sections. This lets the authors estimate event-study style child penalties in places where true panel event studies were previously impossible.
- **What the event studies show.** In most countries, men and women follow roughly parallel pre-trends before parenthood, but women's employment drops sharply after the first child while men's employment barely moves. The qualitative pattern is remarkably pervasive.
- **Main global finding.** Child penalties are widespread but *enormously heterogeneous*. At the continent level, Latin America has the largest average child penalty (about **38%**), Africa the smallest (about **9%**), Asia around **18%**, and Europe, North America, and Oceania around **25–30%**.
- **Within-continent heterogeneity matters.** The cross-country spread inside continents is often larger than the spread across continents. Asia is a striking example: child penalties are close to zero in much of Southeast Asia and China, but extremely large in South Asia, the Middle East, Japan, and South Korea.
- **Development result.** Child penalties explain only a tiny fraction of gender inequality at low levels of development, but become the dominant driver of labour market gender inequality at high levels of development. In advanced economies, eliminating gender inequality is close to eliminating child penalties.
- **Structural transformation interpretation.** The paper argues that child penalties rise as economies move away from subsistence agriculture and toward salaried work in industry and services. The separation between the home and the workplace makes specialization within families more costly and more gendered.
- **Marriage also matters.** In low- and middle-income countries, women often face *marriage penalties* in addition to child penalties. These marriage penalties fade with development, while child penalties become more important. The development process is therefore associated with a substitution from marriage penalties to child penalties.
- **City-level result.** Child penalties are generally larger in cities than at the national level. Even countries with small overall penalties can contain large urban penalties, consistent with urban job structures being less flexible and more specialized.
- **Historical validation.** Using historical census data for today's high-income countries, the paper shows that child penalties used to be very small in the pre-industrial era and rose over

the course of development. This historical evidence lines up closely with the paper’s modern cross-country patterns.

- **Why the paper matters.** The paper does not just document gender gaps. It explains *which part of the gender gap is tied to family formation*, how that part varies around the world, and why it changes with development.

2 Data and measurement (key objects)

2.1 Why this paper can build a global atlas

The paper assembles harmonized, nationally representative microdata from **134 countries**. The combined sample covers about **281 million individual-year observations** and more than **95% of the global population**. The goal is not to study one country in depth, but to put child penalties on a globally comparable footing.

The central outcome is **employment**. The paper focuses on employment because it is observed in most data sources, is measurable in both cross sections and panels, and is tightly connected to labour market gender inequality.

2.2 Cross-sectional data sources

For **127 of the 134 countries**, the paper uses repeated cross-sectional data. The source mix is broad:

- **77 countries** from national censuses, including **75 IPUMS International** samples and separate official data for Tunisia and Angola.
- **16 countries** from the **Luxembourg Income Study (LIS)**.
- **17 countries** from the **Demographic and Health Survey (DHS)**.
- **8 countries** from **EU-SILC**.
- **5 countries** from national **Labour Force Surveys**.
- **4 country-specific sources**: Algeria (MICS), China (1% mini-census), the U.K. (APS), and the U.S. (CPS plus ACS).

Interpretation. The paper’s identifying strategy is built specifically to unlock these cross-sectional data. Without the pseudo-event study design, most of these countries would not be usable for event-study evidence on child penalties.

2.3 Panel data and validation samples

For **7 countries**, the paper has true panel data for the main estimates:

- **Administrative panels**: Austria, Denmark, Norway, Sweden, and Switzerland.
- **National panel surveys**: Australia (HILDA) and Japan (JHPS).

The authors also use extra panel datasets to validate the pseudo-event study approach against true event studies. These include Germany (GSOEP), the U.K. (BHPS), the U.S. (PSID and NLSY), Chile (EPS), South Korea (KLIPS), and Russia (RLMS), in addition to the panel countries listed above.

2.4 Outcome measurement and harmonization

Employment is harmonized as closely as possible across sources. The preferred measure is **current employment status over a short horizon**, usually the previous week or previous 15 days. Where possible, the paper excludes protected parental leave from the employment measure.

In some countries, the only available employment concept is annual employment or employment including paid leave. The authors show that this measurement issue has little effect on the estimated penalties except in a handful of countries with very long paid-leave duration.

Interpretation. The paper is careful not to pretend that all countries measure employment in exactly the same way. Instead, it harmonizes aggressively and then checks whether the remaining inconsistencies materially alter the results.

2.5 Key objects and definitions

Raw gender gap in employment. Let L^g denote the average employment rate for gender $g \in \{w, m\}$. The raw gender gap is

$$\text{Raw Gender Gap} = \frac{L^m - L^w}{L^m}. \quad (1)$$

Event time. The first child birth is indexed to occur at event time $t = 0$. The paper studies outcomes around this event, typically from several years before birth to ten years after birth. The omitted base year is $t = -2$, the year before pregnancy.

Age restrictions. For the event-study analysis, age at first birth is restricted to **20–45**. For raw gender-gap calculations, the broad sample includes people with and without children aged **15–56**.

2.6 Scaled and unscaled child penalties

The paper uses two related concepts:

- **Unscaled child penalty:** the decline in women’s employment relative to men’s in percentage points.
- **Scaled child penalty:** the same decline expressed as a percentage of women’s counterfactual employment absent children.

The scaled penalty is the paper’s main measure because it normalizes by the level of employment women would have had without children, making countries more comparable. The paper later shows that scaled and unscaled penalties are strongly positively related, so the substantive cross-country patterns are not driven mechanically by scaling.

2.7 Historical data

To validate the development patterns over long horizons, the paper also uses historical census data for **seven countries**: Canada, Denmark, Iceland, Norway, Sweden, the U.K., and the U.S. For the U.S., the series is extended to 2020 by combining historical censuses with modern CPS and ACS data.

This historical extension is important because it lets the authors ask whether today's high-income countries used to look like today's low-income countries with respect to family-related gender gaps.

2.8 Descriptive facts that motivate the paper

Before any causal or quasi-causal interpretation, the raw descriptive facts are already striking:

- Gender gaps in employment vary from about **5 percentage points** in the most equal countries to about **90 percentage points** in the least equal countries.
- Direct cross-country comparisons show huge gaps even among nearby countries, for example China versus South Korea or Denmark versus the Czech Republic.
- The relationship between the raw gender gap and GDP per capita is **inverse-U shaped**: the gap first rises with development and later falls.

Interpretation. These descriptive patterns motivate the main question of the paper: how much of this global variation is specifically tied to parenthood and family formation?

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline panel event study

The starting point is the standard event-study design used in the child-penalty literature when true panel data are available. For each gender $g \in \{w, m\}$, the paper estimates:

$$Y_{it}^g = \alpha^g \cdot D_{it}^{Event} + \beta^g \cdot D_{it}^{Age} + \gamma^g \cdot D_{it}^{Year} + \nu_{it}^g. \quad (1)$$

Here:

- Y_{it}^g is the labour market outcome of individual i of gender g at event time t .
- D_{it}^{Event} is the set of event-time dummies around first birth.
- D_{it}^{Age} is a full set of age dummies.
- D_{it}^{Year} is a full set of calendar-year dummies.

Interpretation. The event-time coefficients measure how employment changes around first child birth, net of lifecycle effects and aggregate time effects.

3.2 Scaling the event-study coefficients

The paper scales the estimated event-time coefficients by the predicted employment level absent children:

$$P_t^g = \frac{\hat{\alpha}_t^g}{E[\tilde{Y}_{it}^g | t]}. \quad (2)$$

Here $\tilde{Y}_{it}^g$ is the predicted outcome from equation (1) after omitting the event-time contribution. Thus, P_t^g is the percentage effect of parenthood on employment at event time t .

Interpretation. Scaling makes a 5-point employment loss more consequential in a country where baseline female employment is 20% than in a country where it is 80%.

3.3 Defining the child penalty

The paper then defines the child penalty as the average post-child effect on women relative to men, net of any pre-child difference:

$$\text{Child Penalty} = E[P_t^m - P_t^w | t \geq 0] - E[P_t^m - P_t^w | t < 0]. \quad (3)$$

Interpretation. A positive value means parenthood widens the male-female employment gap. In most countries, this happens because women's employment falls sharply after first birth while men's employment barely changes.

3.4 Pseudo-event study construction

The main methodological challenge is that for individuals observed *without* children in a cross section, we do not know whether or when they will become parents. The paper solves this by matching.

Consider a parent i observed at event time 0 in calendar year y , at age a , with characteristics X_i . The authors create a surrogate pre-child observation by matching this person to a childless individual j observed in year $y - n$, age $a - n$, with the same characteristics $X_j = X_i$. This creates a synthetic observation for event time $t = -n$.

By repeating this for $n = 1, \dots, 5$, the paper reconstructs a pseudo-panel with five years of pre-child data. The resulting pseudo-panel is then used exactly like a regular event-study panel.

Interpretation. The design is not trying to follow the same person back in time. It is trying to build the most comparable pre-child counterfactual profile for someone who is observed as a parent today.

3.5 Matching variables and practical implementation

The matching variables are deliberately chosen to be light but globally available:

- calendar year,
- age,
- gender,

- education (four categories),
- marital status (four categories),
- rural versus urban status.

The paper shows that adding education and marital status is especially important in some countries. These variables remove the positive-selection problem that would otherwise show up as a spurious jump in men's employment at event time zero.

3.6 Validation against true event studies

The authors validate the pseudo-event study approach in two ways.

First validation: male non-event check. Because child birth should be close to a labour-market non-event for men, any positive jump in men's employment at $t = 0$ signals selection bias in the matching. The baseline matching specification passes this check.

Second validation: direct comparison with true panel event studies. In countries where panel data are available, the authors purposely discard the pre-child panel information, reconstruct a pseudo-panel, and then compare pseudo-event-study estimates with true event-study estimates. Across twelve countries, the two sets of child-penalty estimates line up extremely closely, with an R^2 of about **0.96**.

Interpretation. This validation exercise is one of the strongest features of the paper. It shows that the pseudo-event study is not just plausible in theory; it reproduces the panel-data benchmark in practice.

3.7 Fraction of the gender gap explained by child penalties

To connect child penalties to overall gender inequality, the paper uses a simple accounting identity:

$$\text{Fraction Explained} = \frac{\text{Child Penalty} \times \text{Fraction Mothers}}{\text{Raw Gender Gap}}. \quad (4)$$

This formula uses **unscaled** child penalties and **unscaled** gender gaps. It translates the penalty of parenthood into a contribution to the aggregate employment gap.

Interpretation. A country can have a large child penalty but a modest fraction explained if its raw gender gap is already huge before children arrive.

3.8 Joint marriage-child event study

To separate marriage penalties from child penalties, the paper estimates a horse-race specification with event-time dummies for both first marriage and first child:

$$Y_{itt'}^g = \alpha^g \cdot D_{it}^{Child} + \eta^g \cdot D_{it'}^{Marriage} + \beta^g \cdot D_{it}^{Age} + \gamma^g \cdot D_{it}^{Year} + \nu_{itt'}^g. \quad (5)$$

Interpretation. This specification asks: once the timing of marriage is accounted for, how much of the post-family gap is specifically tied to parenthood, and how much to marriage itself?

3.9 Simpler family-penalty specification for broad coverage

Because precise marriage timing is unavailable in all countries, the paper also estimates a simpler specification:

$$Y_{it}^g = \alpha^g Child_i + \eta^g EverMarried_i + \beta^g \cdot D_{it}^{Age} + \gamma^g \cdot D_{it}^{Year} + \nu_{it}^g. \quad (6)$$

This produces country-level estimates of:

- the **child penalty**,
- the **marriage penalty**,
- and the **family penalty**, defined as their sum.

Interpretation. Equation (6) is less granular than equation (5), but it lets the authors study the development path of family penalties across the whole world sample.

3.10 Development decomposition

The paper then studies how the gender gap evolves with GDP per capita by decomposing it into:

- a child-related gap,
- a marriage-related gap,
- and a residual gap.

This decomposition is implemented using LOESS smoothing across GDP per capita. The residual component captures forces such as education differences, norms, or discrimination that are not directly tied to family formation.

Interpretation. This is not a structural decomposition in a deep causal sense. It is an informative accounting exercise that shows how the source composition of gender inequality shifts over development.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimates

The baseline identifying assumption is the same as in the standard child-penalty literature: the *precise timing of first child birth* is orthogonal to non-child changes in women’s labour market outcomes relative to men’s, once age and year effects are controlled for.

In the pseudo-event setting, this assumption is combined with the matching assumption that childless individuals with the same age, year, education, marital status, and rural/urban status provide a credible counterfactual for the pre-child path of those who are observed as parents.

4.2 Why the pseudo-panel is not obviously biased

The main threat is positive selection: women (or men) who become parents may differ from childless matches in ways that also predict employment dynamics. The paper tackles this directly.

The strongest diagnostic is the male event study. If the matching were poor, one would often see men's employment jump at event time zero. In practice, after adding education and marital status to the match, the male series are flat in the relevant countries.

Interpretation. The paper uses men as a built-in placebo. That is a smart and convincing design check.

4.3 Validation against panel benchmarks

The second credibility pillar is the head-to-head comparison with true panel data. The paper shows that the pseudo-event studies and the actual event studies are nearly identical in countries where both can be estimated.

This matters because it converts a potential criticism — “cross sections cannot do event studies” — into an empirical question that the authors answer directly.

4.4 Marriage is a potential confound, and the paper deals with it

Because parenthood often follows marriage very closely, especially in low-income settings, a naive child penalty can partly absorb effects that really occur at marriage. The paper therefore:

- estimates separate marriage event studies,
- estimates joint marriage-child event studies,
- and compares full-sample estimates with samples in which marriage and first birth are more clearly separated in time.

Result: marriage penalties do exist in many countries, but the child-penalty results are not driven by a simple marriage confound.

4.5 Measurement comparability is imperfect but mostly not decisive

Cross-country work always raises the concern that results reflect different measurement conventions. The paper addresses this in three ways:

1. it harmonizes employment as much as possible,
2. it documents where paid leave may affect the measure,
3. and it shows that excluding versus including paid leave changes the results little except in a few very long-leave countries.

Interpretation. The paper is not claiming perfect comparability. It is showing that the main global patterns are too strong to be an artefact of small harmonization differences.

4.6 No instrumental variable in the core atlas

Despite the title of this section, the core design does *not* rely on an instrument. This is important. The paper is not claiming to identify child penalties from exogenous fertility shocks such as twins or sibling sex mix. Instead, it estimates reduced-form event-study effects around the first child.

The paper explicitly positions itself relative to the fertility-IV literature and argues that the atlas has a different purpose: global measurement of child penalties and their relation to gender inequality and development.

4.7 What the paper can and cannot identify

The paper identifies how labour market outcomes diverge around family formation. It does *not* separately identify the deep mechanisms behind those penalties, such as:

- social norms,
- childcare policy,
- occupational flexibility,
- employer discrimination,
- or household bargaining.

Likewise, the structural-transformation interpretation is well supported by the data but remains an economic interpretation of robust reduced-form patterns, not a fully estimated structural model.

4.8 Relation to the literature

The paper sits at the intersection of three literatures:

- the child-penalty literature for high-income countries,
- the fertility and female labour supply literature that often uses IV strategies,
- and the development literature on structural transformation and gender inequality.

Its main value is to connect these literatures with a single globally comparable empirical framework.

5 Tables and figures

5.1 Figure 1 and Appendix Table A.1: data sources and global coverage

Read:

- Figure 1 maps the different underlying data sources used for the 134-country atlas.
- The sample covers roughly **95.5% of the world's population**.

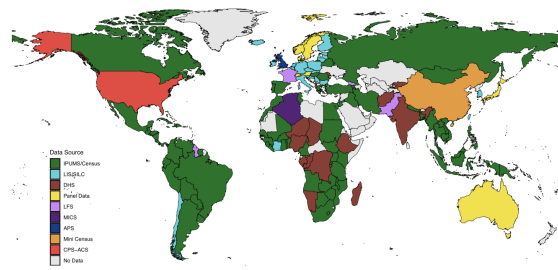


FIGURE 1
Map of data sources

Notes: This map summarizes the data sources used for the 134 countries included in our child penalty atlas. Each colour represents a specific data source. Data for the countries labelled as "Panel Data" refer to either administrative panels or panel surveys. Data for all other countries come from either national censuses or repeated cross-sectional surveys. Appendix Table A.1 provides complete details on the data used for each country.

TABLE A.1: CROSS-COUNTRY MICRO DATA

Country	Source	Years	No. Waves	Employment Measure	Maternity Leave		No. Obs.
					Excluded	Max Days	
Afghanistan	DHS	2015	1	Weekly	Yes	60	28,709
Albania	DHS	2008-2017	2	Weekly	No	330	15,392
Algeria	MICS	2012-2019	4	Weekly	No	91	91,816
Angola	Census	2014	1	Weekly	Yes	56	541,452
Argentina	IPUMS	1970-2001	4	Weekly	Yes	70	3,391,576
Armenia	IPUMS	2001-2011	2	Weekly	No	1,095	189,992
Australia	Panel Data	2001-2019	19	Weekly	No	143	387,695
Austria	Panel Data	1981-2017	38	Annual	No	730	14,104,688
Bangladesh	IPUMS	1991-2011	3	Weekly	No	84	7,250,559
Belarus	IPUMS	1999-2009	2	Weekly	Yes	1,095	660,960
Belgium	LIS	1985-2017	21	Annual	No	344	88,473
Benin	IPUMS	1979-2013	4	Weekly	No	56	536,792
Bolivia	IPUMS	1976-2001	3	Weekly	Yes	30	486,681
Botswana	IPUMS	1991-2011	3	Weekly	No	84	77,961
Brazil	IPUMS	1991-2010	3	Weekly	Yes	183	18,447,111
Bulgaria	SILC	2007-2020	14	Weekly	No	730	58,282
Burkina Faso	IPUMS	1996-2006	2	Weekly	No	70	577,352
Burundi	DHS	2010-2016	2	Weekly	Yes	84	22,320
Cambodia	IPUMS	1998-2008	2	Weekly	No	90	730,506
Cameroon	IPUMS	1976-2005	3	Weekly	No	70	662,598
Canada	IPUMS	2011	1	Weekly	Yes	590	292,388
Chad	DHS	1996-2014	3	Weekly	Yes	70	17,343
Chile	LIS	1990-2017	13	Weekly	No	167	839,202
China	Mini Census	2005	1	Weekly	Yes	152	811,326
Colombia	IPUMS	1973-2005	3	Weekly	Yes	91	2,130,657
Congo, Dem. Rep.	DHS	2007-2013	2	Weekly	Yes	56	23,862
Congo, Rep.	DHS	2005-2011	2	Weekly	No	105	14,376
Costa Rica	IPUMS	1973-2011	4	Weekly	Yes	91	368,480
Cote d'Ivoire	LIS	2002-2015	3	Weekly	No	98	38,799

- Appendix Table A.1 details, country by country, the source, employment measure, coverage, and sample size.

Economic message. The atlas is not built on one survey or one region. It is a truly global data assembly, and the paper's methodological contribution is what makes such broad coverage feasible.

5.2 Figure 2: heatmap of raw gender gaps in employment

Read:

- Raw gender gaps range from about **5pp** in the most equal countries to around **90pp** in the most unequal countries.
- Gaps are relatively small in many high-income countries, East Asia, and Central Africa.
- Gaps are large in Latin America, North Africa, the Middle East, and South Asia.

Economic message. Before turning to child penalties, the paper documents that global gender inequality itself is hugely uneven across space. The later question is how much of that map can be explained by family formation.

5.3 Figure 3: validation of the pseudo-event study

Read:

- For Austria, Chile, Denmark, Japan, and the U.K., pseudo-event studies and true panel event studies are nearly identical.
- Across twelve countries where both are feasible, pseudo-panel and panel child-penalty estimates lie very close to the 45-degree line.
- The reported cross-country validation fit is about $R^2 = 0.96$.

Economic message. The paper's main empirical innovation works. This figure is the credibility

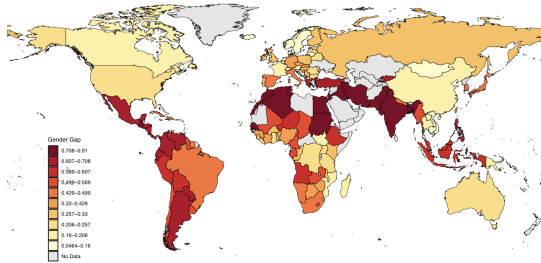


FIGURE 2
Heatmap of gender gaps in employment

Notes: This figure presents a heatmap of raw gender gaps in employment. The gender gap is defined as $(L^m - L^w)/L^m$, where L^g denotes the average employment rate for gender $g = w, m$. The map divides countries into deciles of the gender gap, with darker colours representing larger gaps. The sample includes individuals with and without children between the ages of 15 and 56. For individuals observed with children, we restrict attention to those who had their first child between the ages of 20 and 45.

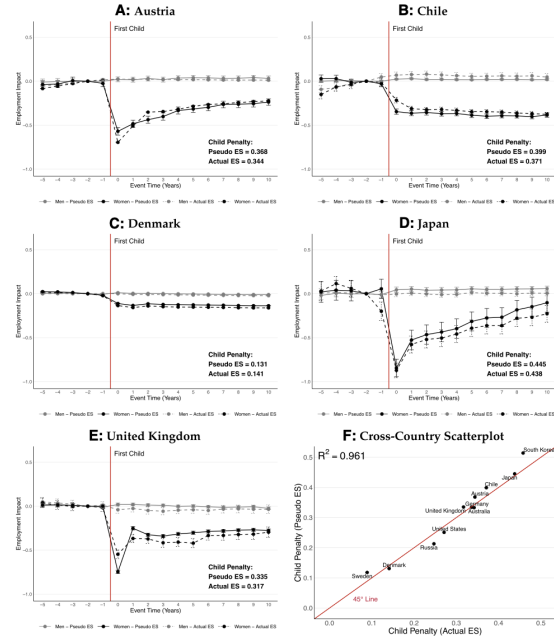


FIGURE 3
Validation of pseudo-event study approach

Notes: This figure validates the pseudo-event study approach against an actual event study approach using panel data. Figure 3A-E consider selected countries, comparing pseudo-event studies of first child birth (solid lines) to actual event studies of first child birth (dashed lines). Each series shows, for men and women, respectively, the percentage impact of child birth on the employment rate at each event time t , i.e. P_t^m and P_t^w defined in equation (2). Each panel also displays average child penalties over event times 0–10 as defined in equation (3). Age at first birth is restricted to 20–45, and the error bars depict 95% confidence intervals based on robust standard errors. Figure 3F presents a scatterplot of child penalties estimated from the two approaches for all countries where the validation is feasible.

backbone of the atlas.

5.4 Figure 4: child penalties by continent

Read:

- In every continent, women's employment drops after first birth while men's series stay roughly flat.
- Latin America has the largest average child penalty at about **38%**.
- Africa has the smallest at about **9%**, while Asia is around **18%** and Europe, North America, and Oceania are around **25–30%**.

Economic message. The child penalty is a global phenomenon, but it is not uniform. Magnitude, not qualitative pattern, is where the world differs.

5.5 Figures 5–8: country examples in Europe, Asia, Latin America, and Africa

Read:

- Europe shows large variation: for example, Denmark is around **14%** while the Czech Republic is around **50%**.
- Asia is even more heterogeneous: Bangladesh is around **62%**, Jordan around **64%**, while Vietnam

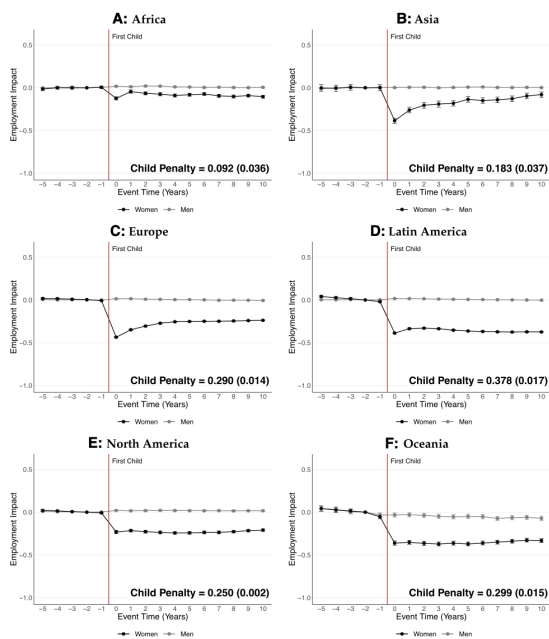


FIGURE 4
Child penalties by continent

Notes: This figure presents event studies of first child birth by continent. Each panel shows an event study of the impact of first child, depicted by the vertical line, on the employment rate of men (grey series) and women (black series). The series depict the percentage impact of child birth at each event time t , i.e. P_t^{PM} and P_t^{PW} defined in equation (2). These coefficients are population-weighted averages of the coefficients estimated for each country in the given continent (among countries included in our data). Each panel also displays the average child penalty over event times 0–10 as defined in equation (3). Age at first birth is restricted to 20–45. The error bars depict 95% confidence intervals based on robust standard errors.

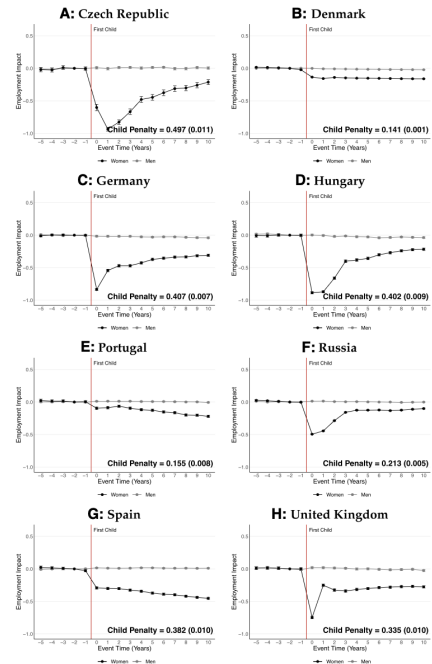


FIGURE 5
Child penalties in European countries

Notes: This figure presents event studies of first child birth for a selection of European countries. Each panel shows an event study of the impact of first child, depicted by the vertical line, on the employment rate of men (grey series) and women (black series). The series depict the percentage impact of child birth at each event time t , i.e. P_t^{PM} and P_t^{PW} defined in equation (2). Each panel also displays the average child penalty over event times 0–10 as defined in equation (3). Age at first birth is restricted to 20–45. The data used for each country are described in Table A.1. The error bars depict 95% confidence intervals based on robust standard errors.

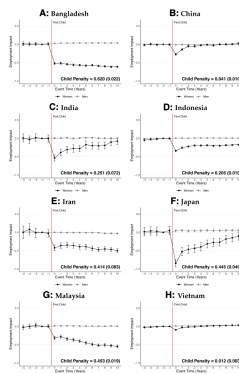


FIGURE 6
Child penalties in Asian countries

Notes: This figure presents event studies of first child birth for a selection of Asian countries. Each panel shows an event study of the impact of first child, depicted by the vertical line, on the employment rate of men (grey series) and women (black series). The series depict the percentage impact of child birth at each event time t , i.e. P_t^{PM} and P_t^{PW} defined in equation (2). Each panel also displays the average child penalty over event times 0–10 as defined in equation (3). Age at first birth is restricted to 20–45. The data used for each country are described in Table A.1. The error bars depict 95% confidence intervals based on robust standard errors.

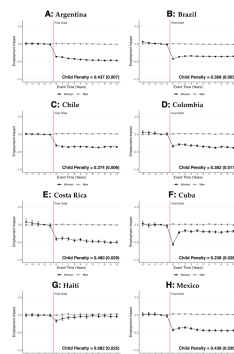


FIGURE 7
Child penalties in Latin American countries

Notes: This figure presents event studies of first child birth for a selection of Latin American countries. Each panel shows an event study of the impact of first child, depicted by the vertical line, on the employment rate of men (grey series) and women (black series). The series depict the percentage impact of child birth at each event time t , i.e. P_t^{PM} and P_t^{PW} defined in equation (2). Each panel also displays the average child penalty over event times 0–10 as defined in equation (3). Age at first birth is restricted to 20–45. The data used for each country are described in Table A.1. The error bars depict 95% confidence intervals based on robust standard errors.

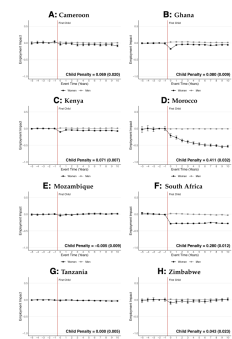


FIGURE 8
Child penalties in African countries

Notes: This figure presents event studies of first child birth for a selection of African countries. Each panel shows an event study of the impact of first child, depicted by the vertical line, on the employment rate of men (grey series) and women (black series). The series depict the percentage impact of child birth at each event time t , i.e. P_t^{PM} and P_t^{PW} defined in equation (2). Each panel also displays the average child penalty over event times 0–10 as defined in equation (3). Age at first birth is restricted to 20–45. The data used for each country are described in Table A.1. The error bars depict 95% confidence intervals based on robust standard errors.

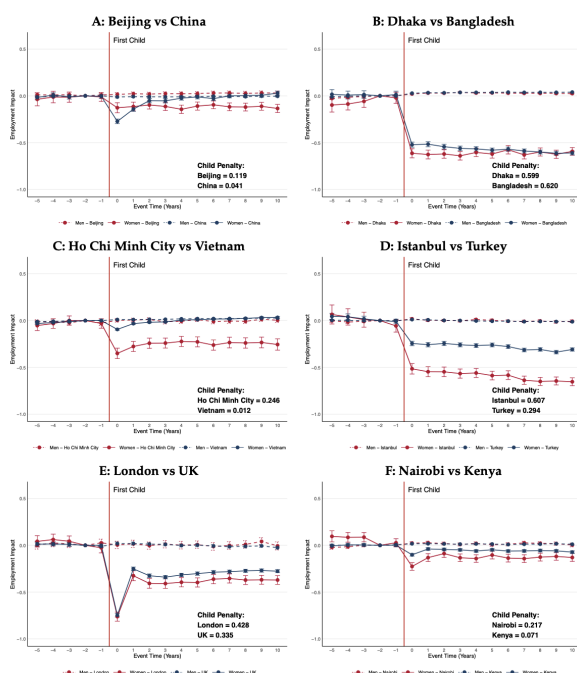
is about 1%.

- Latin America is more uniformly high, with many countries clustered around **35–50%**.
- Africa splits into two groups: many Central African countries have small or zero penalties, while countries such as Morocco and South Africa show large penalties.

Economic message. Geography alone is not destiny. Nearby countries can have very different penalties, suggesting an important role for institutions, labour market structure, and culture.

5.6 Appendix Figure A.6 and Figure 9: city-level penalties and global heatmap

FIGURE A.6: CHILD PENALTIES IN CITIES VS COUNTRIES



Notes: This figure compares event studies of first child birth in large cities (red lines) and countries as a whole (blue lines). Each panel shows event studies of the impact of first child on the employment rate of men (dashed lines) and women (solid lines) in cities and countries, respectively. The series depict the percentage impact of child birth at each event time t , i.e. P_t^C and P_t^N defined in equation (2). Each panel also displays the average child penalty over event times 0-10 as defined in equation (3). Age at first birth is restricted to 20-45. The error bars depict 95% confidence intervals based on robust standard errors. Our pseudo-event study approach yields precise estimates even at the city-level.

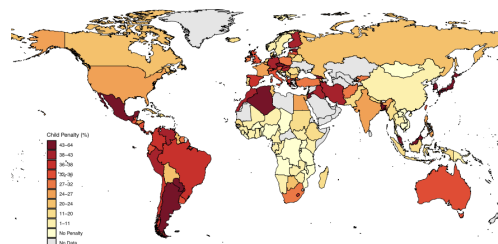


FIGURE 9

Heatmap of child penalties

Notes: This figure presents a heatmap of child penalties in employment, defined in equation (3). The map divides countries into deciles of the child penalty, with darker colours representing larger penalties. Countries with a penalty less than or equal to zero fall in the "No Penalty" category. Countries for which we do not have data fall in the "No Data" category. The event studies of child birth used to estimate child penalties for all 134 countries in our dataset are provided in Appendix Figure A.5.

Read:

- Cities usually exhibit larger child penalties than national averages.
- Examples highlighted in the paper include Beijing **12%** versus China **4%**, Ho Chi Minh City **25%** versus Vietnam **1%**, Istanbul **61%** versus Turkey **29%**, London **43%** versus the U.K. **34%**, and Nairobi **22%** versus Kenya **7%**.
- Figure 9 then aggregates all country estimates into a world heatmap of child penalties.

Economic message. Urban labour markets appear to intensify the child penalty. This foreshadows the paper's later link between penalties, urbanization, and structural transformation.

5.7 Figures 10 and 11: fraction of gender inequality explained and scaled versus unscaled penalties

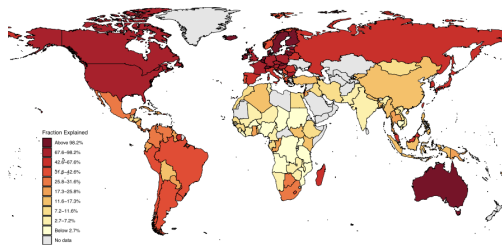


FIGURE 10
Heatmap of fraction of gender gap explained by child penalties

Notes: This figure presents a heatmap of the fraction of the gender gap in employment explained by child penalties. The fraction explained is calculated using equation (4). The map divides countries into deciles of the fraction explained, with darker colours representing larger fractions explained.

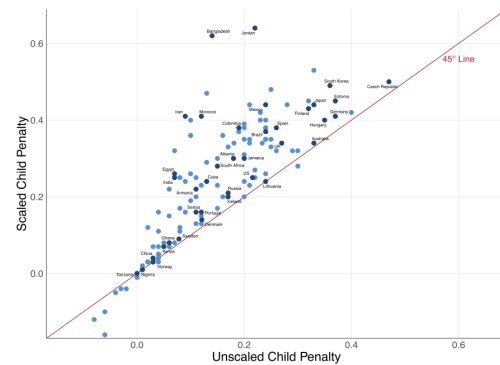


FIGURE 11
Scaled versus unscaled child penalties

Notes: This figure plots scaled child penalties against unscaled child penalties across all 134 countries in our sample. Unscaled child penalties, on the x-axis, represent average effects on the employment rate of women relative to men in absolute terms (pp). Scaled child penalties, on the y-axis, represent average effects on the employment rate of women relative to men in per cent of the counterfactual employment rate absent children, as defined in equations (2)–(3).

Read:

- Figure 10 shows that child penalties explain a much larger share of gender inequality in North America, Europe, and Oceania than in Africa and Asia.
- Figure 11 shows a strong positive relationship between scaled and unscaled child penalties.
- A few Middle Eastern and South Asian countries stand out because they combine large unscaled penalties with very low female employment even absent children.

Economic message. The paper's key comparative statics are not a scaling artefact. What changes with development is not only the level of the penalty but also its share in total inequality.

5.8 Figures 12–14: marriage penalties, family penalties, and development

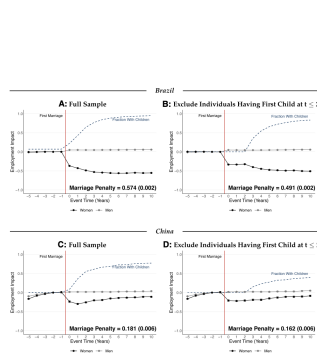


FIGURE 12
Marriage penalties: A case study of Brazil and China

Notes: This figure presents event studies of first marriage in Brazil and China. Each panel shows an event study of the impact of marriage, depicted by the vertical line, on the employment rate of men (grey series) and women (black series). The series depicts the percentage impact of marriage at each event time t , i.e. T^* and T^* defined in equation (3). Each panel also displays the average marriage penalty over event times $t=0$ as defined in equation (5). The fraction of individuals with children at each marriage event time is shown by the dashed blue line. The sample consists of the full sample of females for whom marriage timing is observed (left panels), and the subsample of females who have their first child at least three years after marriage (right panels). Age at first marriage is restricted to 20–45. The error bars (dashed vertical) depict 95% confidence intervals based on robust standard errors.

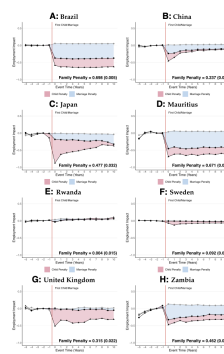


FIGURE 13
Family penalties by country: Joint event studies of first marriage and first child

Notes: This figure presents joint event studies of first marriage and first child in selected countries. These are based on the joint event study specification (3). Each panel shows an event study of the impact of first marriage (blue area) and first child (pink area) on the employment rate of men (grey series) and women (black series). The series depicts the percentage impact of marriage and children, respectively, at each event time t . Each panel also displays an average family penalty—the sum of marriage and child penalty—over event times $t=0$. Age at first marriage and age at first birth are both restricted to 20–45.

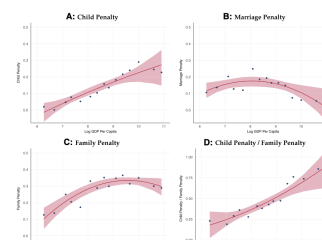


FIGURE 14

Substitution between marriage and child penalties over the course of development

Notes: This figure shows benchmarks of child penalties (Panel A), marriage penalties (Panel B), family penalties (Panel C), and child-to-family penalty ratio (Panel D) by GDP per capita. These are scaled penalties to the employment rate (defined from specification (6)). The family penalty is defined as the sum of child and marriage penalties. Countries have been grouped into 15 equal-sized bins of GDP per capita, and the ratio shows their average across all countries in each bin. Each panel also includes a quadratic fit, depicted by the red line. The GDP data come from the World Bank's WDI for the year 2019.

Read:

- Figure 12 shows that Brazil and China have sizable marriage penalties, though some of the apparent marriage effect reflects children arriving soon after marriage.
- Figure 13 shows stacked joint event studies for selected countries and highlights that family penalties exceed child penalties in many lower-income settings.
- Figure 14 shows that child penalties rise with GDP per capita, marriage penalties are large at low and middle income but fade at high income, and the child-to-family penalty ratio rises from about **25%** to almost **100%**.

Economic message. Family formation always matters, but *which part* of family formation matters changes over development: marriage first, then children.

5.9 Figures 15 and 16: development decomposition and structural transformation

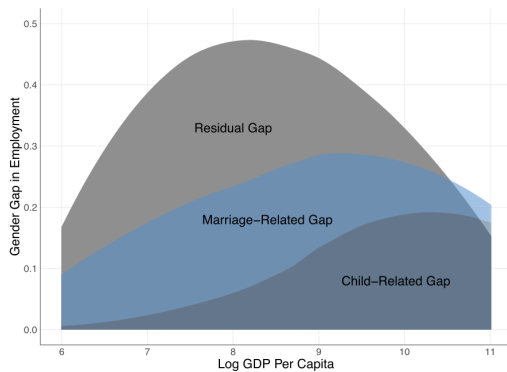


FIGURE 15
Decomposition of gender gap by levels of development

Notes: This figure decomposes the gender gap in employment at each level of GDP per capita into a child-related gender gap (dark blue), a marriage-related gender gap (light blue), and a residual gender gap (grey). The gaps are shown in absolute terms, i.e. the percentage point difference in the employment rates of men and women. The estimates of child and marriage penalties used for the decomposition are obtained from specification (6). The decomposition is implemented using LOESS to smooth the GDP profile of each gender gap—the total gap, the child-related gap, and the marriage-related gap—and calculating the residual gap as the difference between the smoothed total gap and the sum of the smoothed child- and marriage-related gaps. The GDP data come from the World Bank's WDI for the year 2019.

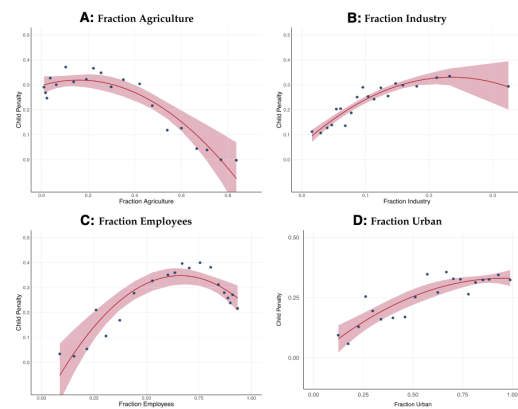


FIGURE 16
Child penalties and structural transformation

Notes: This figure shows bincatters of child penalties against the fraction of the labour force in agriculture (Panel A), the fraction of the labour force in industry (Panel B), the fraction of the labour force in salaried employment (Panel C), and the fraction of the population living in urban areas (Panel D). Each panel includes a quadratic fit, depicted by a red line. The analysis includes all 134 countries in our dataset, and it exploits within-country variation by disaggregating large countries into subnational regions. Countries are disaggregated when (i) the population is greater than 100 million people and (ii) our data include information on region of residence. In each panel, different locations (countries or regions) have been grouped into 20 ventiles of the given development indicator, and the y-axis shows the average child penalty across all locations in each ventile. All development indicators are obtained from the same data and sample as the child penalty estimates.

Read:

- Figure 15 decomposes the gender gap into child-related, marriage-related, and residual components across GDP per capita.
- The total gender gap follows an **inverted-U** pattern over development.
- Child-related inequality rises steadily with development; marriage-related inequality is important early and fades later.
- Figure 16 shows that child penalties are negatively related to agriculture and positively related to industry, salaried work, and urbanization.

Economic message. The central interpretation of the paper is visible here: structural transformation changes the organization of work and family life, and that is what makes child penalties increasingly important.

5.10 Figures 17 and 18: long-run historical evidence

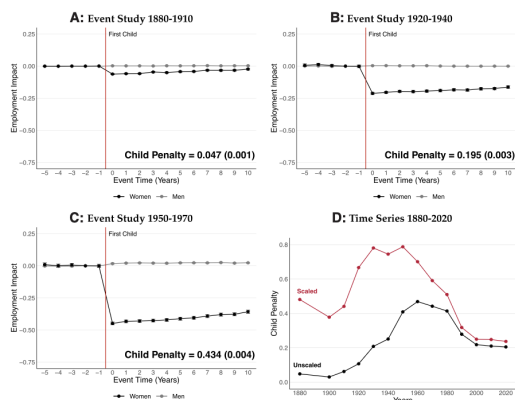


FIGURE 17

Child penalties in the U.S., 1880-2020

Notes: This figure illustrates the evolution of child penalties in the U.S. between 1880 and 2020. Panels A-C present event studies of the impact of first child birth, depicted by the vertical line, on the employment rate of men (grey series) and women (black series) in different historical time periods. Each series depicts the unscaled impact (pp) of child birth at each event time t , i.e. $\hat{\delta}_t^m$ and $\hat{\delta}_t^w$ in equation (1). Each panel also displays the average unscaled child penalty over event times 0-10. Panel D presents the full time series of decade-level child penalties between 1880 and 2020. This panel shows both unscaled penalties (pp) and scaled penalties (%). Age at first birth is restricted to 20-45. The error bars depict 95% confidence intervals based on robust standard errors.

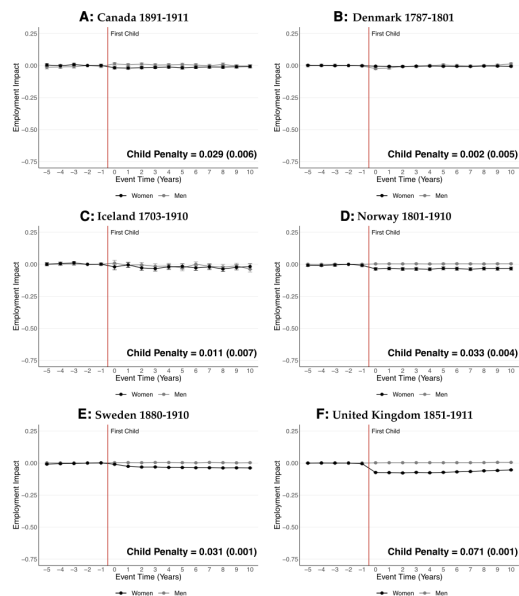


FIGURE 18

Historical child penalties in other countries

Notes: This figure presents event studies of first child birth using historical data from different high-income countries. Each panel shows an event study of the impact of first child, depicted by the vertical line, on the employment rate of men (grey series) and women (black series). The series depict the unscaled impact (pp) of child birth at each event time t , i.e. $\hat{\delta}_t^m$ and $\hat{\delta}_t^w$ in equation (1). Each panel also displays the average unscaled child penalty over event times 0-10. Age at first birth is restricted to 20-45. The error bars depict 95% confidence intervals based on robust standard errors.

Read:

- In the United States, unscaled child penalties were under **5pp** in the late nineteenth century, rose to almost **50pp** around the mid-twentieth century, then declined to about **20pp** and later plateaued.
- Historical event studies for Canada, Denmark, Iceland, Norway, Sweden, and the U.K. show child penalties that are either zero or very small in the pre-industrial era.
- The historical series line up closely with the modern cross-country patterns by development level.

Economic message. Today's high-income countries once looked like today's low-income countries in terms of family-related gender inequality. This is powerful evidence for the development interpretation.

6 Short summary

- The paper builds a global atlas of child penalties in employment using harmonized microdata from 134 countries.
- Its main empirical innovation is a pseudo-event study that reconstructs pre-child observations from repeated cross sections.
- The pseudo-event approach is strongly validated against true panel event studies.
- Most countries display a sharp and persistent employment drop for women after the first child, while men's employment is largely unaffected.

- Child penalties vary enormously around the world: Latin America is high on average, Africa low, and Asia and Europe highly heterogeneous.
- Child penalties explain only a small fraction of gender inequality at low levels of development, but become the dominant driver at high levels of development.
- Marriage penalties matter especially in low- and middle-income countries, but fade with development.
- Child penalties are closely linked to structural transformation: they rise as economies urbanize and shift from agriculture toward salaried work in industry and services.
- Historical evidence from today's rich countries confirms that child penalties were very small in earlier development stages and rose over time.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that family formation is a central and globally variable source of labour market gender inequality. Child penalties are nearly universal in pattern but highly heterogeneous in magnitude. Their importance rises systematically with development, to the point that in high-income economies they account for almost all remaining gender inequality in employment.

7.2 Economic intuition

At low levels of development, much work is organized in ways that are closer to the household, especially in agriculture and self-employment, so parenthood does not generate large additional labour market penalties for women. As economies develop, work becomes more formal, more urban, more salaried, and more spatially separated from home production. That makes specialization inside the family more consequential. The result is that children increasingly widen the labour market gap between women and men.

7.3 Takeaway

The broad lesson is that gender inequality should not be studied only through average wage or employment gaps. A large and development-dependent part of the story is about what happens when people form families. The paper's atlas shows that understanding global gender inequality requires understanding how marriage, parenthood, and labour market structure interact.